

Jeffery Liu

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Penultimate-year Mechatronics Engineering student specialising in mechanical design, with hands-on electrical and controls experience. Real design, build and test experience beyond coursework: an R&D internship at Holmes Solutions where I designed an apparatus upgrade that was delivered and tested and co-authored the resulting technical paper, over three years running a licensed vehicle dealership I founded, and performance vehicle work from tuning consultation to a turbocharged engine rebuilt in my garage. Love for all types of performance vehicles.

EDUCATION

University of Canterbury, BE(Hons) Mechatronics Engineering,

2024 – present

- Third year, expected graduation 2027. Second-year grade average: A-.
- Offered two competitive UC entrance international scholarships (each around 10 awarded nationally); not taken after gaining residency.
- Relevant coursework: solid mechanics, computational mechanical analysis, dynamics, fluid mechanics, controls & vibrations, power electronics, embedded systems, engineering economics.

EXPERIENCE

Mechatronics Engineering Intern, Holmes Solutions

Nov 2025 – Feb 2026

Roadside & Marine sector, R&D consultancy (453.5 hours)

- Investigated premature mooring rope failures end to end: researched the cause and designed an interchangeable sleeve upgrade handling loads above 50 tonnes, delivering test data toward safer mooring guidance industry-wide.
- Mapped apparatus constraints and verified the design with hand calculations and SolidWorks FEA. Flagged a non-conservative capacity check on the revised load case with my own calculations and escalated it for structural review.
- Produced construction-ready drawings and a staged build plan, then sourced contractor quotes and defined scope, sequencing and resourcing with the contractor before preparing weld joints on site. Also supported a second intern's crash-test bogie build and roadside full-scale crash-test setup, gaining workshop fabrication and testing experiences on site.
- Design was built and passed full-scale testing in 2026; co-authored the resulting technical paper, presented at an international marine conference in August 2026.

Reference: Ray Keane, Senior Engineer, ray.keane@holmessolutions.com

Founder & Licensed Vehicle Dealer, Jeff's Auto Retail

2023 – 2026

- Registered as a sole trader at 18, set up my own workshop as site of operation and prioritised fast turnover.
- Sourced, diagnosed, repaired and sold used vehicles end to end, and prepared vehicles to WOF standard.
- Bought and sold 35+ cars solo. Handled all the business sides including pricing, listings, customer relations, bookkeeping, tax and compliance.

Junior Vehicle Dynamics Engineer, UC Motorsport (FSAE)

Apr 2026 – present

Junior lead on the tyre-temperature sensing system project

- Researched and specified system requirements, selecting an IR pyrometer for real-time onboard tyre temperature logging, intended to feed a variable friction coefficient into the car's traction control system.
- Owned schematic, PCB layout (KiCAD) and component selection for a Stage 1 standalone ESP32-based sensor module, then self-taught SMT and assembled the board using a pick and place machine and reflow.
- Wrote firmware in Circuit Python with USB mass storage to make data retrieval behave like using a USB stick.
- Housing that clips on the car and must handle vibration and up to 2.5 G designed and 3D printed. First track testing done, more testing incoming to further validate the system capability.

Performance Consultant & Tuning Support, Custom Culture & Tuning

2025 – present

- Consult directly with clients; quote required parts, repairs and achievable power levels from hardware limits, then work with an Auckland-based tuner where I handle datalogging and mechanical adjustments for tuning.

Reference: Amit Lal, Tuner/Business Owner, cct.tune@gmail.com

Bicycle Technician & Maintenance Tutor, Adventure Cycles

2019 – 2023

- Diagnosed and repaired complete systems (including drivetrain, brake and suspension) across all bike types.
- Ran monthly public maintenance tutoring workshops for customers and applied the same skills at school through my TravelWise leadership role. I also taught other young bike mechanics.

Bike Mechanic, Evo Cycles / Pitcrew Cycles**Oct 2021 – Jan 2023**

- Assembled bikes and carried out mechanical servicing during school holidays.

Workshop Apprentice, Steelguard Ltd**Mar 2023 – Aug 2023**

- Assisted during a post-flood company relocation. Gained machine shop experience. Practised engineering drawing interpretation, sanding, and MIG welding.

PROJECTS

Turbocharged Engine Rebuild & Swap, 2006 Subaru Legacy GT**2025 – present**

- Diagnosed an internal engine failure from running symptoms and a compression test, then chose to build a stronger engine rather than repair the original.
- Tore down the engine and had the block and heads machined, then rebuilt it myself with stronger internals (bearings, rings, head studs), upgraded cooling and oil systems, and an upgraded turbo and supporting bolt-ons to match my power goal.
- Validated the build through datalogging, monitoring and fault-checking it myself and working with my tuner to refine the calibration from logged data; now daily driven, with motorsport-style tune/calibration planned.

Line Follower Robot, Team University Project**2025**

- Lead designed and manufactured the complete mechanical platform and drivetrain in SolidWorks for a robot built from scratch by the team (custom PCB, sensors and motor control). Contributed to PID tuning.
- Casted silicone tyres for maximum traction with 3D printed mould.
- Fine-tuned weight distribution to minimise skid drag during acceleration and maximise passive braking from front skid contact. Implemented rubber band braking system that could be adjusted from number of twists.
- Set up private testing in my garage with our replica track to protect design and avoid testing congestion.
- Set a practice lap quick enough to have won the event; finished 3rd, 0.14s off first due to late motor failure

Pick and Place CNC Machine, Team University Project**2025**

- Team of five designed and built a pick-and-place CNC machine, CAD modelled in SolidWorks, targeting a sub-\$200 build assembled at home without specialist tools for hobbyists. Used a waterjet-cut aluminium plate architecture.
- Built belt-end drive across the X and Y axis and leadscrew for Z axis, traced a binding fault during assembly to belt misalignment. I ran the bulk of assembly and disassembly, learning DFMA through practise.
- Delivered for \$128.85 against target, assembled in 1 hour, position error within 0.3 mm.

RoboCup Autonomous Robot, Team University Project**2026 – present**

- Team of three building a fully autonomous robot for a timed arena challenge: navigating and avoiding obstacles, collecting target weights and ejecting dummies, and returning them to base using homing logic.
- Leading mechanical design and working across electrical and embedded systems; proposed the design concept the team chose to build. Build in progress: ramp and drum intake through multiple iterations, ToF sensing linked to drive control for obstacle response, and a sorting mechanism now in development.

SKILLS

Mechanical & production (through internship, vehicle related experiences, and coursework):

Vehicle systems, diagnostics & repair, fabrication, MIG & arc welding, plasma cutting, oxy torch, machine tool operation (drill, mill, lathe), structural verification, CAD (SolidWorks incl. FEA, Fusion 360), engineering drawing, GD&T, CAM (3D printing, laser cutting, CNC), MATLAB & Simulink, COMSOL, dynamic testing setup, DFMA.

Electronics & hardware (through vehicle related experiences, UC Motorsport, and coursework):

Electrical diagnosis, soldering, auto & marine wiring, PCB design (KiCAD), LTSpice, SMT, hardware testing, power electronics design.

Controls & embedded (through UC Motorsport, tuning support, and coursework):

PLC (ladder logic) programming, embedded software (C, Python), firmware development, control theory, performance ECU tuning, datalogging.

Business & communication (through vehicle dealership, internship, group projects, and coursework):

Small business management, sales, bilingual (English/Mandarin), marketing (social media), consultation, contractor coordination, technical report, literature review.

ACTIVITIES

Petrol head, boating, fishing, camping, snowboarding, cooking.